Supplementary Information for

Aggregate biocapacity does not identify its composition: a two-land model of ecological-capital drawdown

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This document supplements the main text (referred to as "the paper"). Section S1 gives the symbol table and stated extensions; S2 the derivations and proofs behind Propositions 1–3 and the macro-ratio identities; S3 the numerical protocol and verification record; S4 extended model results (yield sweep, drawdown-depth ladder, delay and restoration checks, recovery metric and gate-sign phase dependence, regime table); S5 the empirical details (NFA Edition 2026 endpoints and component accounts, the same-edition sensitivity, the exploratory crop-proxy Divisia calculation and ϕ_0-robustness, observation-operator limits, regeneration anchors, and the unpopulated physical-stock scoreboard specification); S6 the policy extensions; S7 the one-stock comparator; S8 the controlled technology-wave/no-wave comparison. Equation numbers without the prefix S refer to the paper.

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S1 Symbol table and stated extensions

Following GFN/NFA practice, each annual footprint or biocapacity account value is reported in gha, not gha yr^{-1} (Global Footprint Network, 2009, 2019). With continuous model time measured in years, B, E, Y_f, Y_c, S have units gha; their time derivatives have units gha yr^{-1}. The leaky debt state D is the time-integral of the shortfall and has units gha yr, so $\dot{D}$ and ηD are gha. This internal model liability is not an NFA account variable.

Table S1: Symbols, meanings, units and roles. Per-hectare annual account yields are in gha ha^{-1}; $b_{G,c}$ is a regeneration-value coefficient in gha yr ha^{-1}; annual biocapacity, footprint and shortfall account amounts are in gha; the leaky debt state is in gha yr.

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	per-hectare capital-land quality (capacity index)	dimensionless	state
P	human population	cap	state
D	leaky accumulated ecological debt (time-integral of shortfall)	gha yr	state
b_f	fast-land account yield per hectare, $(b_{f0} + T_b)e^{-\alpha D}$	gha ha^{-1}	derived
$\tilde{b}_f(t)$	OWID/FAOSTAT cereal yield, a physical productivity sentinel (not all crops)	tonnes ha^{-1}	empirical input
$\hat{b}_f(t)$	Base-anchored index-scaled proxy yield used only in SI S5	gha ha^{-1}	proxy construction
b_{f0}	base fast-land yield per hectare	gha ha^{-1}	parameter
b_c	capital-land direct yield	gha ha^{-1}	parameter
$b_{G,c}$	capital regeneration-value coefficient	gha yr ha^{-1}	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1 - q/q_{\max})$	yr^{-1}	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\mathrm{eff}}$)	gha ha^{-1}	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	gha ha^{-1}	derived
Y_f, Y_c	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha	derived

Symbol	Meaning	Unit	Role
B	model biocapacity $Y_f + Y_c$; or NFA total under its account convention	gha	aggregate
E_{prod}	NFA Global Ecological Footprint of Production	gha	empirical account
E	footprint account value, eP (or frozen)	gha	aggregate
$X,\ C$	crop-proxy account $X = \hat{b}_f A_f$ and residual $C = B - X$; neither is physical capital	gha	illustrative accounting split
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\text{conv}}$	$\text{ha}\,\text{yr}^{-1}$	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	$\text{gha}\,\text{cap}^{-1}$	parameter
r	intrinsic population growth rate	yr^{-1}	parameter
ρ_c	quality regeneration rate	yr^{-1}	parameter
$q_{\max}$	quality ceiling	dimensionless	parameter
μ	cropland maintenance rate (reserve→fast)	yr^{-1}	parameter
η_f	fast-land retirement rate (fast→reserve)	yr^{-1}	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt	$\text{gha}^{-1}\,\text{yr}^{-1}$	parameter
η	debt-repayment rate	yr^{-1}	parameter
$\tau_g,\ \tau_p$	regeneration and demographic delays	yr	parameter
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	dimensionless	parameter
R_{rc}	surplus-gated restoration, $\chi_r A_r\,[B - E]_+$ (reserve→capital)	$\text{ha}\,\text{yr}^{-1}$	gated flow
χ_r	restoration coefficient	$\text{gha}^{-1}\,\text{yr}^{-1}$	parameter
T_b	technology yield increment, $\omega_{Tb}\Delta b/(1 + e^{-\kappa_w(t - t_{wave})})$	$\text{gha}\,\text{ha}^{-1}$	off except in SI S8
ω_{Tb}	activation flag for the technology wave	dimensionless	0 baseline; 1 in SI S8
Δb, κ_w, t_{wave}	amplitude, steepness and center of the wave	$\text{gha}\,\text{ha}^{-1}$; yr^{-1}; yr	active only in SI S8
$A_c^{\min}$, $A_r^{\min}$, $A_f^{\max}$, $K_{\min}$	capital viability floor, reserve floor, arable ceiling, carrying-capacity floor	ha, ha, ha, cap	typed floors
A_{tot}	total bioproductive area	ha	parameter
k	$\mu/(\mu + \eta_f)$, fast share of non-capital land at equilibrium	dimensionless	derived
$B_{eq}(A_c)$	frozen-demand balance branch, $b_{f0} k A_{tot} - (b_{f0} k - b_c) A_c$	gha	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	gha	derived
$\lambda_{1,2}$	deficit-side Jacobian eigenvalues	yr^{-1}	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A = R_B/\psi_f$)	dimensionless	diagnostics
ϕ_0, w_X, w_C	base-year cropland share and log-mean weights	dimensionless	empirical decomposition
$s_X, \bar{s}_X, \bar{s}_C$	current and average fast-book shares	dimensionless	empirical decomposition
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}})$	dimensionless	derived

Symbol	Meaning	Unit	Role
R_{fc}, χ	fast→capital restoration variant, $\chi A_f [B-E]_+$	$\mathrm{ha\,yr^{-1}}$; $\mathrm{gha^{-1}\,yr^{-1}}$	extension (off)
L_c, H_c	timber liquidation flow $[H_c - \sigma_c Y_c]_+ / b_{G,c}$ and its target	$\mathrm{ha\,yr^{-1}}$; gha	extension (off)

Extension status. (i) A timber-liquidation channel $L_c = [H_c - \sigma_c Y_c]_+ / b_{G,c}$ removes capital area when an annual timber-account target H_c exceeds the harvestable capital account amount; $\gamma = 1/b_{G,c}$ ($\mathrm{ha\,gha^{-1}\,yr^{-1}}$) applies only to this channel and remains off. (ii) A direct fast-to-capital restoration $R_{fc} = \chi A_f [B-E]_+$ (surplus-gated) can replace or supplement R_{rc} and remains off. (iii) A reservation $\sigma_f, \sigma_c < 1$ withholds a share of each flow from demand (Section S6) and remains off. (iv) The technology wave T_b of Eq. (14) is off in the baseline, main figures and tables, and recovery-gate experiment; it is activated for the controlled matched comparison in SI §S8. The separate wave-on recovery case mentioned in S4.4 is a solver-regression check only and is excluded from the reported scenario set. The v44 solver propagates the wave consistently through current B, the returned account series, and delayed carrying-capacity history. (v) An endogenous per-capita footprint $e(t)$ is an offered extension not implemented here.

Conventions. The positive part is exact; conversion is a rate through τ_{conv}; land is conserved one-to-one across A_f, A_c, A_r while q is a separate capacity book; annual NFA account amounts B, E, Y_f, Y_c, S are in gha, while the leaky time-integral D is in gha yr and $\dot{D}, \eta D$ are in gha; a sustainable fast book exists because of the maintenance term μA_r. The reserve floor $A_r^{\min}$ and arable ceiling $A_f^{\max}$ are declared domain constraints that are checked after each run (never violated in any reported run: the minimum reserve in the drawdown sweeps is 0.80 ha) rather than enforced by clamps; $A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$ and $K_{\min}$ are enforced.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiating $B = b_f A_f + \Upsilon_c(q) A_c$ with $b_f = (b_{f0} + T_b) e^{-\alpha D}$ and substituting Eqs. (6)–(7) gives the general identity

$$\dot{B} = \Delta b_{\mathrm{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c} G_c'(q) A_c\, \dot{q} + \Upsilon_c R_{rc} - \alpha \dot{D}\, b_f A_f + \frac{\dot{T}_b}{b_{f0} + T_b}\, b_f A_f. \quad \text{(S1)}$$

The conversion term has the sign of Δb_{conv}; the quality term has the sign of $G_c'(q)$ (non-negative for $q \leq q_{\max}/2$, non-positive above it, since $\dot{q} \geq 0$) and is bounded in magnitude by $b_{G,c} \rho_c \dot{q} A_c$, which is small at the baseline; the debt term is non-positive while debt accrues; the technology term is non-negative while the wave rises; the restoration term is non-negative. Proposition 1 of the paper is the case $T_b \equiv 0$, $R_{rc} \equiv 0$, D fixed. In the demonstration run D is not fixed (it rises to 0.20 gha yr and decays), so the debt term is present, and the mask holds nevertheless: the conversion term dominates by two orders of magnitude at the start ($\Delta b_{\mathrm{conv}} u_c \approx 0.78\,\mathrm{gha\,ha^{-1}} \times 0.27\,\mathrm{ha\,yr^{-1}} = 0.21\,\mathrm{gha\,yr^{-1}}$ against $b_f(\mu A_r - \eta_f A_f) = -0.005\,\mathrm{gha\,yr^{-1}}$ and $\alpha \dot{D} b_f A_f \approx 0.004\,\mathrm{gha\,yr^{-1}}$).

S2.2 The deficit-side affine system and its spectrum

On the deficit side with $q = q_{\max}$, D frozen, $R_{rc} = 0$ and $S/b_f < A_c - A_c^{\min}$, Eq. (5) gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$, and with $A_r = A_{tot} - A_f - A_c$ Eqs. (6)–(7) become

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\text{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\text{conv}}} \end{pmatrix}. \tag{S2}$$

Hence $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\text{conv}}$ with $k = \mu/(\mu + \eta_f)$. At the baseline $\operatorname{tr} J = -1.0512\,\text{yr}^{-1}$ and $\det J = 0.05353\,\text{yr}^{-2}$, and the eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537\,\text{yr}^{-1},\ -0.9975\,\text{yr}^{-1}. \tag{S3}$$

Because J contains neither E nor the state, the spectrum is constant along the frozen-demand branch (verified at $E/E_{\text{ceil}} = 0.50, 0.90, 0.99, 0.999$: $\lambda_1 = -0.05366\,\text{yr}^{-1}$ in every case). J contains no ρ_c, so λ_1 is exactly independent of the regeneration rate ($-0.0537\,\text{yr}^{-1}$ at $\rho_c = 0.04, 0.08, 0.16\,\text{yr}^{-1}$). It depends mildly on τ_{conv} (-0.0553, -0.0537, -0.0505, $-0.0417\,\text{yr}^{-1}$ at $\tau_{\text{conv}} = 0.5, 1, 2, 5\,\text{yr}$) and more strongly on the maintenance rate (-0.0254, -0.0537, $-0.1099\,\text{yr}^{-1}$ at $\mu = 0.03, 0.06, 0.12\,\text{yr}^{-1}$); its value should therefore be reported as an order-of-magnitude negative number that never crosses zero, while the qualitative statements (no positive eigenvalue, no fold, no critical slowing) hold across all sweeps. The fixed point of the affine system is $A_c^*(E)$ of Eq. (19) with $A_f^* = k(A_{tot} - A_c^*)$; for $E > E_{\text{ceil}}$ it lies below $A_c^{\min}$ and the trajectory approaches the face $A_c = A_c^{\min}$ asymptotically, with limiting $u_c = 0$ and persistent shortfall $S^* = E - E_{\text{ceil}}$. Two verified members of the no-conversion family at the baseline are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914\,\text{ha}, 0.824\,\text{ha}, 0.762\,\text{ha}, 1.488\,\text{cap})$ and $(1.091\,\text{ha}, 0.500\,\text{ha}, 0.909\,\text{ha}, 1.731\,\text{cap})$, both with $A_r^*/A_f^* = \eta_f/\mu = 0.8333$ exactly; the second is a sustainable cropland equilibrium supported by the maintenance term. The slow mode is the equilibration of the reserve share $A_r/A_f \to \eta_f/\mu$, which is why the strict-inequality window of the demonstration run extends to $\approx 150\,\text{yr}$ although most of the conversion is complete within 25 yr.

In the other branch of the minimum ($S/b_f \geq A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\text{conv}}$ and the distance to the floor obeys $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\text{conv}}$: it decays exponentially and never reaches zero in finite time. This is the reason the floor is approached asymptotically in Fig. 1c of the paper and the operational floor-proximity thresholds depend on the tolerance used to declare proximity (Section S4.2).

S2.3 Characteristic structure of the full delay system

With state $z = (A_f, A_c, q, P, D)$ the linearized system about any point of a given active set is $\dot{z}(t) = J_0 z(t) + J_g z(t - \tau_g) + J_p z(t - \tau_p)$ with characteristic equation $\det \Delta(\lambda) = 0$, $\Delta(\lambda) = \lambda I - J_0 - J_g e^{-\lambda \tau_g} - J_p e^{-\lambda \tau_p}$. J_0 carries the conversion and debt feedbacks on the current state (the 2×2 block above, plus the D row $\partial \dot{D}/\partial z = (-b_f, -\Upsilon_c, -b_{G,c} G_c' A_c, e, -\eta)$ on the deficit side and the $\partial b_f/\partial D = -\alpha b_f$ couplings); J_g has the single entry $G_c'(q^*)$ in the q row; J_p has the row $(rP^{*2}/K^{*2})\, e^{-1}\, \partial B/\partial z$ in the P position (from $\partial \dot{P}/\partial K = rP^2/K^2$ and $K = B/e$). Two scalar delayed modes decouple at the baseline. The quality mode linearized at $q_{\max}$ is $\dot{\xi} = -\rho_c \xi(t - \tau_g)$ (rate $-0.08\,\text{yr}^{-1}$ without the delay); its characteristic root satisfies $\lambda = -\rho_c e^{-\lambda \tau_g}$, and since $\rho_c \tau_g = 1.6 > \pi/2$ the linear delayed mode would be oscillatory. In the nonlinear model this cannot occur because $\dot{q} = G_c(q(t - \tau_g)) \geq 0$ for $q(t - \tau_g) \in [0, q_{\max}]$ and q is clamped at $q_{\max}$, so quality is monotone non-decreasing and converges to $q_{\max}$; the linear oscillation is an artifact of linearizing a one-sided law at its boundary. The population mode on the no-conversion face is the delayed logistic law with $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold, so it is non-oscillatory at the baseline. No entry, determinant sign or eigenvalue is inherited from the one-stock characteristic

equation of Section S7; each is derived per active set (no conversion; active conversion; floor face $A_c = A_c^{\min}$; $K = K_{\min}$).

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\text{ceil}}$ the limiting capital state is $A_c = A_c^{\min}$, approached asymptotically, and the persistent shortfall feeds the debt. With $B_{eq}(A_c^{\min}; D) = b_{f0}e^{-\alpha D}k(A_{tot} - A_c^{\min}) + b_c A_c^{\min}$, the debt equation $\dot{D} = E - B_{eq}(A_c^{\min}; D) - \eta D$ has the fixed point $D^*(E)$ solving $\eta D = E - B_{eq}(A_c^{\min}; D)$; the right-hand side is increasing in D, so the fixed point is unique and attracting whenever $\eta > \alpha b_{f0} k(A_{tot} - A_c^{\min})$ ($0.034\,\text{yr}^{-1} < 0.05\,\text{yr}^{-1}$ at the baseline). Solving numerically, $D^* = 1.186\,\text{gha yr}$ at $E = 1.158\,\text{gha}$ and $D^* = 5.375\,\text{gha yr}$ at $E = 1.238\,\text{gha}$, with $B(D^*) = 1.099$ and $0.969\,\text{gha}$ and persistent shortfalls $S^* = 0.059$ and $0.269\,\text{gha}$; the integrated debt values quoted in earlier versions (1.21 and 5.39 gha yr) agree to within 2% (finite horizon). Under frozen demand the debt therefore saturates; it is unbounded only if the shortfall is.

S2.5 Boundedness asymmetry and the crossover shortfall

From Eq. (12), $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The first term lies in $[\ln b_{f0},\ \ln(b_{f0} + \Delta b)]$, so the technology channel contributes at most $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ to $\ln b_f$ at the baseline. From Eq. (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S}$ the running supremum of the shortfall, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ and the degradation channel contributes at least $-\alpha\bar{S}/\eta$. The two are equal when $\bar{S} = S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296\,\text{gha}$ at the baseline. For the fast-book yield itself, $Y_f = b_f A_f$ shares this structure; the capital term $\Upsilon_c A_c$ is unaffected by either channel. Under the additive law $b_f = b_{f0} + T_b - \alpha D$, both channels are bounded quantities of the same kind (Δb against αD^*) and the logarithmic asymmetry disappears, which is why the paper states the result for the multiplicative form.

S2.6 Macro-ratio identities

Define $R_B = E/B$, $R_A = E/Y_f$ and the fast-land share $\psi_f = Y_f/B$; then $R_A = R_B/\psi_f$ is an identity. The direct-yield share inside the capital book, $\psi_c = b_c A_c/Y_c$, is a different quantity and $R_A = R_B/\psi_c$ is false. $R_B = 1$ is aggregate balance. With baseline shares $\sigma_f = \sigma_c = 1$, $R_A > 1$ (demand exceeds the fast-book account yield) is necessary but not sufficient for conversion, because the shortfall requires $E > B = Y_f + Y_c$, not merely $E > Y_f$; with reserved flows the threshold changes. The capital book declines iff $u_c > R_{rc}$, with u_c given by Eq. (5). Under the baseline shares, active conversion therefore requires $R_B > 1$; in the stated reservation extension $\sigma_f, \sigma_c < 1$, conversion and a declining capital book can occur while $R_B \le 1$, so aggregate overshoot is not required in that extension. "$R_B = 1$ is the fold" belongs to the one-stock frozen-demand reducer (Section S7), not to the two-land model. At the regeneration maximum $q^* = q_{\max}/2$ the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at the baseline; this is a property of that point, not a structural constant.

S3 Numerical protocol and verification record

Solver. Method of steps with an explicit first-order (Euler) update; exact positive part (a smooth ramp leaks conversion when $E < B$); clamps $A_c \ge A_c^{\min}$, $A_f \ge 0$, $q \in [0, q_{\max}]$, $P \ge 0$, $D \ge 0$, $K \ge K_{\min}$; the reserve is computed as $A_r = A_{tot} - A_f - A_c$ after each update and the conservation residual $|A_f + A_c + A_r - A_{tot}|$ is recorded at every step (maximum 2.2×10^{-16} ha in the baseline runs). The pre-run histories of B (for $K(t - \tau_p)$) and q (for $G_c(q(t - \tau_g))$) are constant at the initial state over the full interval $[-\max(\tau_g, \tau_p), 0)$; $B(0)$ is calculated from the model, never inserted as zero. The v44 solver also propagates the active logistic T_b into current B, returned output and delayed B history. In the earlier solver, a zero B sample at $t = 0$ caused an artificial dip in delayed carrying capacity, and the active wave was omitted from returned and delayed

B; both bookkeeping errors are corrected. Step sizes: $\Delta t = 0.05\,\mathrm{yr}$ for the demonstration run and the bisection thresholds; $\Delta t = 0.1\,\mathrm{yr}$ for the drawdown-depth sweeps (checked against 0.05 at selected points; the operational initial-population floor-proximity threshold at $A_{c0} = 1.0\,\mathrm{ha}$ moves from 2.955 to 2.98 cap, which is why thresholds are reported as bands).

Delayed versus current quality in the regeneration term. The deposited solver (`work/twoland_fixed.py`) computes the delayed quality $q(t - \tau_g)$ but evaluates the update of q at the current quality. Table S2 compares the two variants at the baseline; because q is monotone and the regenerative term is small ($b_{G,c}G_c \leq 0.016\,\mathrm{gha\,ha^{-1}}$ against $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$), the differences are below 1% in every reported quantity. The corrected variant (`work/twoland_fixed_delayed.py`, a one-line change) is deposited alongside; all quantities in the paper are quoted to a precision at which the two agree.

Table S2: Effect of evaluating the regeneration term at the delayed rather than the current quality (baseline registry). Biocapacity B, fast-book contribution Y_f, and capital contribution $\Upsilon_c(q)A_c$ are in gha; land areas A_c, A_f, A_{c0} are in ha; the operational initial-population floor-proximity threshold P_0 is in cap; delays and step intervals are in yr.

Quantity	current q (deposited)	delayed $q(t-\tau_g)$
Demonstration window with $\dot{B} > 0$, $\dot{A}_c < 0$ (yr)	150.3	149.0
Mask-end values B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Capital contribution $\Upsilon_c(q)A_c$ (initial $\to$ mask end, gha)	0.08364 $\to$ 0.04038	0.08364 $\to$ 0.04038
Fast-book contribution Y_f (initial $\to$ mask end, gha)	0.51000 $\to$ 0.78462	0.51000 $\to$ 0.78462
P_0 floor threshold ($A_{c0} = 1.0\,\mathrm{ha}$, $\tau_g = 5\,\mathrm{yr}$, $\Delta t = 0.1\,\mathrm{yr}$)	2.955	2.955
P_0 floor threshold ($A_{c0} = 1.0\,\mathrm{ha}$, $\tau_g = 20\,\mathrm{yr}$)	2.955	2.954
P_0 floor threshold ($A_{c0} = 1.0\,\mathrm{ha}$, $\tau_g = 40\,\mathrm{yr}$)	2.955	2.954

Recompute-then-assert. The figure script (`make_fig_model.py`) regenerates the demonstration run and asserts $B(0) = 0.594\,\mathrm{gha}$; at the end of the simultaneous $\dot{B} > 0$, $\dot{A}_c < 0$ window, $B = 0.825\,\mathrm{gha}$, $A_c : 1.500 \to 0.808\,\mathrm{ha}$, $A_f : 0.600 \to 0.923\,\mathrm{ha}$, and the capital contribution $\Upsilon_c(q)A_c : 0.08364 \to 0.04038\,\mathrm{gha}$ (a 51.7% decline); the fast-book contribution $Y_f = B - \Upsilon_c(q)A_c$ rises from 0.51000 to 0.78462 gha. It also checks a window of $150 \pm 2\,\mathrm{yr}$ and an area-conservation residual $< 10^{-12}\,\mathrm{ha}$; it recomputes dimensionless k, the slope $b_{f0}k - b_c$ in $\mathrm{gha\,ha^{-1}}$, $E_{\mathrm{ceil}} = 1.1384\,\mathrm{gha}$ and the five balance points in ha, and asserts each against the stated value before the figure is written. The drawdown sweep (`drawdown_depth.py`) writes `si_drawdown_depth.csv`, from which Fig. 1c and Tables S4–S5 are produced. Eigenvalues and ceilings in Table 2a of the paper are computed from the closed-form J of Section S2.2 and agree with the numerical Jacobian of `work/verify_tab_conv_basin.py` to four decimals.

Well-posedness. The state space $\Omega = \{(A_f, A_c, A_r, q, P, D) : A_i \geq 0,\ q \in [0, q_{\max}],\ A_f + A_c + A_r = A_{tot},\ K \geq K_{\min}\}$ is forward invariant under the clamped scheme; the only non-smoothness is the exact positive part at $S = 0$ and the minimum in Eq. (5), both of which are Lipschitz, so solutions exist and are unique for each history. Stability statements at $S = 0$ are one-sided (deficit side).

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 extends Table 2a of the paper with the slope of the balance branch. The sign of Δb_{conv} is robust over a threefold range of the standing value: at $b_f = 0.85\,\mathrm{gha\,ha^{-1}}$,

$\Upsilon_c(q^*) = 0.058, 0.066, 0.074\,\mathrm{gha\,ha^{-1}}$ for $b_{G,c} = 0.4, 0.8, 1.2\,\mathrm{gha\,yr\,ha^{-1}}$, so $\Delta b_{\mathrm{conv}} = +0.792, +0.784, +0.776\,\mathrm{gha\,ha^{-1}}$.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f (gha ha^{-1})	$\Delta b_{\mathrm{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$ (gha ha^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 gives the minimum capital land reached during endogenous-demand runs ($A_{f0} = 0.6$, $q_0 = 0.9$, $\tau_g = 20$, $\tau_p = 25\,\mathrm{yr}$, $\chi_r = 0.10$, $\Delta t = 0.1\,\mathrm{yr}$, $T = 400\,\mathrm{yr}$) on a grid of initial population P_0; the full grid (step 0.1) is in `si_drawdown_depth.csv`. The depth is continuous and monotone in P_0 and in A_{c0}. The initial population P_0 (cap) at which the minimum comes within 10^{-2}, 10^{-3} and 10^{-4} ha of the floor is 1.8, 1.9, 2.1 ($A_{c0} = 0.40\,\mathrm{ha}$); 2.2, 2.3, 2.4 ($A_{c0} = 0.70\,\mathrm{ha}$); 2.6, 2.7, 2.8 ($A_{c0} = 1.00\,\mathrm{ha}$); 2.8, 2.9, 3.0 ($A_{c0} = 1.18\,\mathrm{ha}$); the bisection with a 10^{-6} ha floor-proximity tolerance ($\Delta t = 0.05\,\mathrm{yr}$, $T = 1000\,\mathrm{yr}$) gives 2.41, 2.70, 2.98, 3.14 cap. The thresholds are identical to three decimals for $\chi_r = 0.10$ and $\chi_r = 0$ (2.387 and 2.955 cap at $A_{c0} = 0.40$ and 1.00 ha, $\Delta t = 0.1\,\mathrm{yr}$), and for $\tau_g = 5, 20, 40\,\mathrm{yr}$ (Table S2). The minimum is reached at the moment the shortfall closes ($B \geq E$ for the first time), which occurs between 3 and 25 yr after the start depending on P_0.

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows, ha) and initial population P_0 (columns, cap); typed floor $A_c^{\min} = 0.05\,\mathrm{ha}$.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1.0	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

S4.3 Recovery metric and the role of the restoration flow

The recovery metric is $\mathcal{R}_c(T) = (A_c(T) - A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}} - A_c^{\mathrm{deg}})$, with $\mathcal{R}_c = 0$ no recovery, 1 full recovery to the reference, < 0 continued decline and > 1 over-restoration. Table S5 evaluates it on the drawdown runs with $A_{c0} = 1.0\,\mathrm{ha}$ ($A_c^{\mathrm{init}} = 1.0\,\mathrm{ha}$, $A_c^{\mathrm{deg}} = \min_t A_c$ in ha, $T = 400\,\mathrm{yr}$): without restoration the capital book stays at its minimum (no return flow exists), with the surplus-gated flow it recovers partially, and at the mildest shortfall it over-restores ($\mathcal{R}_c = 1.28$), moving up the equilibrium family to a higher-capital, lower-biocapacity member while the supported population re-equilibrates downward.

S4.4 Gate-sign phase dependence

The gate-sign comparison is evaluated from a non-equilibrium post-crash state ($A_c = 0.058\,\mathrm{ha}$ near the floor, $A_f = 0.6\,\mathrm{ha}$, dimensionless $q = 0.9$, $P = 1.2\,\mathrm{cap}$) rather than from the degraded equilibrium itself, because at the degraded $R_B = 1$ equilibrium $B = E$ and neither $[B - E]_+$ nor $[E - B]_+$ can fire. The reference is the healthy member of the equilibrium family reached from $A_{c0} = 0.9\,\mathrm{ha}$ ($A_c^{\mathrm{init}} = 0.8225\,\mathrm{ha}$, $P^{\mathrm{init}} = 1.489\,\mathrm{cap}$) and the degraded level is $A_c^{\mathrm{deg}} = 0.0513\,\mathrm{ha}$.

Table S5: Recovery of the capital book after a demand-driven drawdown ($A_{c0} = 1.0$ ha, $A_{f0} = 0.6$ ha, $q_0 = 0.9$, $T = 400$ yr): initial population P_0 is in cap, initial footprint E_0 in gha, capital-land values in ha, and the recovery metric is dimensionless.

P_0	E_0	$\min_t A_c$	t(min) (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	1.076	1.28
2.0	1.100	0.398	3.8	0.398	0.828	0.71
2.5	1.375	0.075	4.0	0.075	0.628	0.60
3.0	1.650	0.050	13.8	0.050	0.540	0.52
3.5	1.925	0.050	20.6	0.050	0.504	0.48
4.0	2.200	0.050	25.0	0.050	0.491	0.46

All rows use the base model ($T_b = 0$). Table S6 reproduces the record (package-local validation script `work/recovery_metric_and_tau_p_scan.py`).

Table S6: Restoration policy and gate sign: recovery metric at $T = 100$ and 900 yr, end-state books, population relative to the reference, extremes over the run and peak debt D (gha yr). Annual account values $B = E$ are in gha; land-book areas are in ha; population ratios and recovery metrics are dimensionless. Every row re-equilibrates to $R_B = 1$ ($E = B$); conservation holds to machine precision in every row.

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.057	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.098	0.765	0.637	0.705	0.86	0.602	1.098	0.546
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

Reading. No restoration gives no recovery; the degraded state is a valid $R_B = 1$ equilibrium with a larger-than-reference population supported by fast land. Surplus-gated restoration recovers the capital book but at $\chi_r \geq 0.05$ restores it beyond the reference because the flow keeps running while $B > E$; the supported population falls modestly ($P/P^{\text{init}} = 0.97$–0.90). A target cap converts this into an exact return ($\mathcal{R}_c = 1.00$, $P/P^{\text{init}} = 1.00$). Deficit-gated restoration is the one gate that acts during the shortfall but is swamped by concurrent conversion at these rates ($\mathcal{R}_c \approx 0$); it is necessary but insufficient unless sized up or paired with a conversion cap. Freezing conversion under surplus-gated restoration raises the peak debt ($0.152 \to 0.546$) because the fast-land release channel can no longer close the shortfall while restoration over-supplies the lower-yield book; A_f falls under the freeze because restoration drains the reserve that maintains it. The earlier v43 wave-on recovery outcome that drove A_c to about 2.44 ha, reduced A_r to 0.027 ha (below its 0.10 ha floor), and drove returned B to 0.150 gha was not a valid model result: that solver applied T_b to the within-step yield but omitted it from returned B and the delayed B history. As a separate solver-regression check, not a reported scenario, the v44 bookkeeping correction applied to the same wave-on recovery initial state ($\Delta t = 0.05$ yr, $T = 900$ yr) gives $A_c^{\max} = 1.731$ ha, $A_r^{\min} = 0.321$ ha, and final $B = 0.899$ gha; the reserve floor is respected. The gate-sign table above uses the base model with $T_b = 0$. The separate controlled wave/no-wave debt comparison is reported in SI §S8.

S4.5 Regime table

Table S7: The composition diagnostic Δb_{conv} (gha ha^{-1}) as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behavior
composition-premium	$\Delta b_{\text{conv}} > 0$	the conversion contribution to accounting B is positive; net B also depends on other flows (the mask is possible in the baseline)
composition-neutral	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at once; no mask; the liquidation sign is clear

S5 Empirical details

S5.1 Data and vintages

The empirical backbone is the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). Although the workbook covers 1961–2025, the primary analysis uses complete world observations for 1961–2023; the 2024–2025 estimated rows are excluded. The sensitivity window, 1961–2022, uses the same edition and same population series, not an earlier-vintage endpoint. The release recalculates common historical years and reports biocapacity by fishing grounds, built-up land, cropland, grazing land, and forest biocapacity. These are account categories, not a common-boundary physical-stock inventory.

The workbook's `pop_1000_persons` field is UN PopStat population in thousands of persons; it is converted to persons before recomputing per-capita values. The field `efp_total_gha` is the Global Ecological Footprint of Production, denoted E_{prod} here. Per-capita values are ratios of the unrounded account totals and population, not the workbook's rounded per-capita columns. Table S8 summarizes endpoint changes. For 1961–2023, B is 9,725,417,562 to 12,142,843,961 gha, population is 3.06487 to 8.09173 billion, B/P is 3.17319 to 1.50065 gha person^{-1}, and E_{prod}/P is 2.19700 to 2.56462 gha person^{-1}. The identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ holds to rounding; it is an extensive/intensive accounting split, not a causal decomposition. The first year in which $E_{\text{prod}}/B > 1$ is 1973 in both windows.

Table S8: World endpoint changes in NFA Edition 2026. The 1961–2022 column is a same-edition sensitivity; 2024–2025 estimates are excluded. Population and per-capita quantities are recomputed from workbook totals.

Measure	1961–2023	1961–2022
$\Delta B/B_{1961}$	+24.8568%	+24.5450%
$\Delta \ln B$	+0.221997	+0.219497
$\Delta P/P_{1961}$	+164.0154%	+161.7211%
$\Delta \ln P$	+0.970837	+0.962109
$\Delta(B/P)/(B/P)_{1961}$	−52.7085%	−52.4131%
$\Delta \ln(B/P)$	−0.748840	−0.742612
$\Delta(E_{\text{prod}}/P)/(E_{\text{prod}}/P)_{1961}$	+16.7327%	+17.6994%
First year $E_{\text{prod}}/B > 1$	1973	1973

S5.2 Divisia decomposition, ϕ_0-robustness and the level fallback

The crop proxy is a deliberately exploratory accounting construction. Let A_f be the independently sourced OWID/FAOSTAT world cropland-area series (arable plus permanent-crop area, ha) and let $\tilde{b}_f$ be the independently sourced cereal-yield series (tonnes per hectare) (Our World in

Data, 2026a,b). Cereal yield is a sentinel for crop productivity, not a measure of all-crop output. The spliced World Bank all-crops production-index series previously examined in the repository is not used because its discontinuous base-period changes make it unsuitable for this decomposition. For a declared 1961 account-share anchor ϕ_0, define $\hat{b}_f(t) = \hat{b}_f(1961)\tilde{b}_f(t)/\tilde{b}_f(1961)$, where $\hat{b}_f(1961)A_f(1961) = \phi_0 B(1961)$, then $X_t = \hat{b}_f(t)A_f(t)$ and $C_t = B_t - X_t$. Thus X is an index-scaled account proxy and C is the residual required to recover the NFA aggregate; C is not observed ecological capital, non-cropland biocapacity, or the model quantity $\Upsilon_c(q)A_c$.

When X and C are positive at both endpoints, the exact log-mean Divisia identity is

$$\Delta \ln B = w_X\left(\Delta \ln \tilde{b}_f + \Delta \ln A_f\right) + w_C \Delta \ln C, \qquad w_X = \frac{L(X_T, X_0)}{L(B_T, B_0)}, \quad w_C = \frac{L(C_T, C_0)}{L(B_T, B_0)}, \tag{S4}$$

where $L(a, b) = (b - a)/(\ln b - \ln a)$ is the logarithmic mean, with its continuous limit at $a = b$. The measured NFA cropland share in 1961 is $\phi_0 = 0.1286472202$ and is used as the central anchor, but the mapping of that account share to the external area–yield proxy is an assumption, not an independently identified physical-capital observation. Table S9 reports the full selected sensitivity for 1961–2023. At the central anchor, the cereal-yield, area, and residual terms are $+0.268662$, $+0.037710$, and -0.084375, summing to $\Delta \ln B = +0.221997$. The same-edition 1961–2022 sensitivity gives $+0.264696$, $+0.037506$, and -0.082705, summing to $+0.219497$. The decomposition is exact for the defined proxy but its allocation depends on ϕ_0 and must not be read as a physical-stock attribution.

Table S9: Sensitivity of the proxy Divisia decomposition to the 1961 account-share anchor ϕ_0 (world, 1961–2023). Contributions and weights are dimensionless. The $\phi_0 = 0.128647$ row uses the measured NFA cropland share in 1961; other rows are illustrative calibrations.

ϕ_0	w_X	w_C	Cereal yield	Area	Residual C	Sum
0.100000	0.1833	0.7957	+0.2088	+0.0293	−0.0162	+0.2220
0.128647	0.2358	0.7352	+0.2687	+0.0377	−0.0844	+0.2220
0.150000	0.2749	0.6893	+0.3133	+0.0440	−0.1352	+0.2220
0.190000	0.3483	0.6008	+0.3968	+0.0557	−0.2305	+0.2220
0.250000	0.4582	0.4581	+0.5221	+0.0733	−0.3734	+0.2220
0.300000	0.5499	0.3178	+0.6265	+0.0879	−0.4924	+0.2220
0.320000	0.5866	0.2454	+0.6683	+0.0938	−0.5401	+0.2220
0.340000	0.6232	0.1009	+0.7100	+0.0997	−0.5877	+0.2220

For $\phi_0 \geq 0.340531$ in 2023 (threshold 0.343400 for 2022), the residual C is nonpositive at the terminal endpoint and logarithmic weighting is invalid. In that range the exact level identity $\Delta B = \Delta X + \Delta C$ is used instead; it remains an accounting identity and carries no stock interpretation. The selected anchor has $X_{2023}/B_{2023} = 0.377784$ and a positive residual, but the proxy fast-book share and residual contribution change materially across the sensitivity values. These calculations use the same NFA B in all rows; the residual absorbs the difference between the index-scaled crop proxy and that aggregate, including account-category, boundary, valuation, vintage, and calibration differences.

S5.3 Cropland-book calibration

Direct NFA cropland-account movements are reported separately from the exploratory proxy. From 1961 to 2023, NFA cropland biocapacity rises 2.870055-fold while NFA cropland area rises 1.198029-fold; the quotient of account biocapacity by NFA area rises 2.395647-fold. For the same-edition 1961–2022 sensitivity the ratios are 2.851441, 1.198326, and 2.379520. The per-hectare quantity is an NFA account-value quotient, not an independent physical crop-yield measure. By comparison, the external OWID/FAOSTAT area series rises 1.173414-fold and cereal yield rises 3.124667-fold over 1961–2023. Those independent series provide a sentinel proxy, not a reconstruction of the NFA cropland account or a measure of all-crops output. The model's

representative $b_{f0} = 0.85\,\text{gha}\,\text{ha}^{-1}$ remains a conditional parameter, not an empirical fit to these account ratios.

S5.4 Observation-operator hierarchy

- *NFA accounting observations.* Total and component biocapacity values in gha support statements about the composition and movement of the NFA accounts under the stated edition and conventions. They do not directly observe biomass, soil condition, wetland extent, fish biomass, or the model state A_c.

- *Fast-provisioning area.* The external world cropland-area series is arable plus permanent-crop area, 1961–2023 (ha) (Our World in Data, 2026a). It is used as an independently sourced proxy input, not treated as identical to the NFA cropland-area series.

- *Crop-productivity sentinel.* The OWID/FAOSTAT cereal-yield series, 1961–2024, is measured in tonnes per hectare (Our World in Data, 2026b). It is not an all-crops index and is not converted into a physical stock. The base-anchored $\hat{b}_f$ is only the index-scaled term in the illustrative account proxy.

- *Ecological-capital components.* Forest extent, growing stock or biomass, soil carbon and condition, wetland extent and condition, and species- or stock-specific fish biomass could inform separate physical observation operators. They would require harmonized spatial boundaries, reference years, definitions, uncertainty, and explicit treatment of overlap. No comparable global physical-stock inventory for the relevant components is assembled here. Existing NFA forest biocapacity and the residual C are not substitutes for those measurements.

- *Model mapping.* In the model, $Y_c = \Upsilon_c(q)A_c$ is a model-defined capital contribution. Empirically, neither A_c nor q is identified by B or by an NFA component; the proxy residual $C = B - X$ is not assumed to equal Y_c. A defensible observation operator would need independent physical measures and a declared mapping before a model-to-data comparison.

  Table S10 summarizes this mapping without equating an NFA category to a model state.

Table S10: Conceptual mapping from two-land model quantities to NFA information and independent proxy inputs. The rows specify possible observation operators, not identities between NFA categories and model states. Annual account values are in gha; land areas are in ha.

Model quantity	Closest account or proxy information	Interpretation and identification limit
Fast area and proxy account A_f, $X = \hat{b}_f A_f$	NFA cropland account; independently sourced OWID cropland area and cereal yield	NFA cropland biocapacity is an account value; the external area–yield product is an illustrative, base-anchored proxy. Neither is an observed model fast-land yield or a one-to-one physical measure.
Capital area, quality and contribution A_c, q, $Y_c = \Upsilon_c(q)A_c$	No direct NFA counterpart; the proxy residual $C = B - X$ is defined by subtraction	In the model $Y_c = \Upsilon_c(q)A_c$; empirically C is not assumed equal to Y_c. It also absorbs differences in boundaries, account categories (including built-up land), valuation, vintage and proxy calibration, and is not a physical-stock measure.
Reserve area A_r	No direct NFA account counterpart	A_r closes the model's conserved-area ledger; it is not recoverable from the NFA aggregate.

Table S11: Observability ladder: what is directly measured, what is identifiable by construction, and what remains unidentified.

Tier	Quantities	Status
Observable physical area input	OWID cropland area [ha]; other mapped land-cover classes where available	Measured extent, but mapping to effective A_c is not direct
Observable population	P [persons]	Direct in the UN PopStat workbook field after unit conversion
Observable productivity sentinel	Cereal yield [tonnes ha^{-1}]	Physical yield of cereals only; not all-crops output or ecological capital
Observable annual account value	NFA B, E_{prod} and components [gha]	Measured only under the NFA accounting convention; value-weighted accounts
Proxy account and residual	$X = \hat{b}_f A_f$ and $C = B - X$ [gha]	Identified by construction for the stated inputs and anchor; not physical stocks and not assumed equal to $\Upsilon_c(q)A_c$
Effective capital contribution	$Y_c = \Upsilon_c(q)A_c$ [gha]	A model quantity; no common-boundary physical observation operator is assembled here
Not separately identifiable	A_c, q, b_c, $b_{G,c}$, ρ_c and D	Requires independent physical-state, quality, regeneration or degradation observations and a declared mapping

S5.5 Regeneration anchors

The capital-land regeneration timescale is a lumped recruitment-response timescale; the anchors bound a band and do not calibrate ρ_c. A lagged saturating recovery model fitted to plot-level age–above-ground-biomass records of Neotropical secondary forests (2ndFOR; Poorter et al., 2016, 1,334 plots, 41 chronosequences, 35 good fits) gives a median $t_{50} \approx 30\,\text{yr}$ and a recovery timescale $\tau_r \approx 38\,\text{yr}$; the same model fitted to deplete-then-rebuild trajectories of collapsed fish stocks (RAM Legacy B/B_{MSY}, 33 episodes) gives $t_{50} \approx 5.6\,\text{yr}$ and $\tau_r \approx 3.4\,\text{yr}$ (cf. Hutchings and Reynolds, 2004; Neubauer et al., 2013); soil organic carbon re-equilibrates after ≈ 17–$23\,\text{yr}$ following land-use change (Poeplau et al., 2011). Forest biomass recovers to about half of old-growth values by $\approx 20\,\text{yr}$ and to 90% at a median $\approx 66\,\text{yr}$ (Poorter et al., 2016) (the 66–95 yr tail lies beyond the core band); only $\approx 29\%$ of collapsed fish stocks recovered to 50% within 5–15 yr (Hutchings and Reynolds, 2004). The two fitted legs bracket the effective timescale (forest slow, fishery fast) with soil between them; the band is ≈ 25–$33\,\text{yr}$ (representative $\approx 29\,\text{yr}$; order 10–40 yr). These are statements about the regeneration rate, not the delay τ_g. In the one-stock comparator the forest value lies at or above that model's ≈ 18–$20\,\text{yr}$ cliff; in the two-land model the drawdown boundary is a demand threshold and the anchors bound a band rather than locate a cliff.

S5.6 Recursive identifiability in the land-type accounts

NFA Edition 2026 shows why the aggregate should be read together with its account composition, without translating account movement into physical-stock movement. From 1961 to 2023, cropland contributes +24.0577 percentage points of the +24.8568% change in total B; built-up land contributes +7.4416 points, forest biocapacity −5.9525, grazing land −0.2688, and fishing grounds −0.4212. The non-cropland account sum, which includes built-up land, changes by only +0.7990 points (1.0092×) because those unlike account movements net against one another. In the same-edition 1961–2022 sensitivity, the corresponding contributions are cropland +23.8183, built-up land +7.2854, forest biocapacity −5.8137, grazing land −0.2890, and fishing grounds −0.4559 points; the non-cropland sum changes by +0.7267 points (1.0083×). These are accounting attributions, not evidence that forest biomass, soil condition, wetland extent, fish biomass, or modeled A_c declined. Built-up land is an NFA account class, not ecological capital; its inclusion in the non-cropland sum is arithmetic, not ecological aggregation.

S5.7 Physical-stock scoreboard: specification, not inventory

No common-boundary global physical-stock inventory for the relevant components has been assembled in this paper, and no physical-stock decline finding or composite is presented. This limitation does not imply that the required source data could never be assembled. The scoreboard in Table S12 is a specification for a future, separately sourced assessment only; it contains no empirical endpoint values and is not a calibration of the model.

Heterogeneous physical stocks cannot be integrated by simply summing their native units: forest biomass or soil carbon (mass), wetland or cropland extent (area), fish biomass (stock by species or stock), freshwater storage (volume), and condition indicators (often dimensionless or categorical) are neither dimensionally commensurate nor necessarily spatially disjoint. Annual flows such as harvest, catch, increment, recharge, or emissions must also remain distinct from stocks and extents. Even after normalization into a common score, overlap, baselines, uncertainty, ecological thresholds, valuation, and permitted substitution determine the interpretation. Component measures therefore form a non-additive vector unless common spatial and temporal boundaries, normalization or valuation, and substitution rules are declared. A scalar composite is not neutral and cannot be inferred from the NFA gha accounts. This distinction is consistent with ecosystem-extent and ecosystem-condition accounts being reported separately from

ecosystem-service and monetary presentations in SEEA Ecosystem Accounting (United Nations et al., 2021).

Table S12: Physical-stock scoreboard specification only. No common-boundary global inventory or endpoint values are assembled in this paper. The listed stocks, extents, condition measures, and annual flows are non-additive.

Component	Physical stock, extent, or condition to define	Annual flow or pressure (separate)	Comparability rule required	Status here
Forests	Extent (ha); growing stock or biomass (m^3 or tonnes C); stated condition	Net annual increment, harvest, or deforestation	Common forest definition, mapped area, reference year, and treatment of plantations	Not assembled
Soils	Soil organic-carbon stock or density for a stated depth; structure/erosion condition	Carbon accumulation/loss or erosion where measured	Consistent depth, bulk density, land cover, and sampling method	Not assembled
Wetlands	Extent (ha) and explicit ecological-condition measure	Carbon accumulation, water storage change, or nutrient flux	Wetland definition and overlap with forest and other land-cover classes	Not assembled
Fisheries	Spawning-stock biomass by species or stock (tonnes)	Recruitment, catch, and landings	Stock identity, assessment method, reference point, and uncertainty	Not assembled
Freshwater	Surface-water or groundwater storage (m^3), level, and quality	Recharge and withdrawal ($m^3\,yr^{-1}$)	Basin or aquifer boundary and separation of stored volume from annual flow	Not assembled

In the two-land model, A_c is an effective area state (ha) and q is a separate capacity-quality index. Neither is directly observed in the NFA aggregate. A physical-stock assessment could inform a model observation operator only after its boundaries and mapping to those effective states are specified. The exploratory residual $C = B - X = B - \hat{b}_f A_f$ is not ecological capital, a measured non-cropland stock, or an estimate of $\Upsilon_c(q)A_c$; the NFA components remain account results rather than physical-stock observations.

S6 Policy extensions

Half-Earth and reservation. The Half-Earth proposal (Wilson, 2016) caps human-usable annual biocapacity account value (gha) at half of the total. In the two-land setting this is a reservation $E \le \sigma B$ with dimensionless σ the human-available share, enforced either as a cap on the fast book or on the carrying capacity; with $\sigma_f, \sigma_c < 1$ in Eq. (4) the shortfall is computed against the reserved flows, so conversion begins earlier for a given demand and the ceiling of Proposition 2 is lowered to σ-weighted values. On the nominal path a reservation that keeps $E \le \sigma B$ succeeds with no liquidation. This is a nominal-path consequence until it is certified against a declared disturbance class (persistent productivity, b_f or ρ_c shocks) under a preregistered retention rule (a governance module is retained only if it improves a declared protection-and-supply score), with the model's declared fit defects converted into an erosion margin on the closed loop's contraction rate; until then it is a property of the nominal calibration, not a robustness claim.

Typed floors and freeze conversion. Reserve $A_r \ge A_r^{\min}$, capital viability $A_c \ge A_c^{\min}$, arable ceiling $A_f \le A_f^{\max}$ and $K \ge K_{\min}$ are separate constraints. Freezing conversion (capping A_f at $A_f^{\max}$ or flooring A_c at $A_c^{\min}$) is the single operator that guarantees $A_c \ge A_c^{\min}$; the conversion rate saturates as $A_c \to A_c^{\min}$ by construction of Eq. (5). A policy that observes only the aggregate B

cannot guarantee capital-land integrity: the failure is informational (aggregation relative to the typed floor), not dynamical.

Institutional signature. In the mobilizing/protective taxonomy of the delay companion (Abaee, 2026a), freeze-conversion and surplus-gated restoration are protective; the mobilizing sign (demand responding to a perceived shortfall by converting more) is the destabilizing one; deficit-gated restoration is not unambiguously protective because it can deepen the short-run shortfall (Section S4.4).

Weak and strong sustainability reading. In the material-cycle reading of the ledger companion (Abaee, 2026c), weak sustainability is the idealized regime in which substitution and regeneration jointly redistribute matter so that it is used as it arises – substitution admissible only through an identified physical pathway, never merely because a weighted aggregate permits it – and strong sustainability is "the regime in which that closure fails – either because no identified physical pathway re-routes the extracted matter, or because depletion outruns the rate at which such a pathway could be deployed", so that a separately binding floor on a critical stock binds. The composite annual account value B (gha) is the weak-sustainability index: the linear, perfectly substitutable member of the aggregator family of the assessment companion (Lemma A of Abaee, 2026b); it can satisfy its aggregate floor while the typed floor on A_c is violated by exactly the amount that the fast-land yield gains against the capital loss, and in that family no positive-elasticity aggregator is uniformly safe (its Theorem S2(ii)). Under exact-tube semantics the intermediate passage during which A_c falls is itself a violation even if the endpoint recovers.

Reporting rule. A growth headline for total biocapacity carries no information about whether cropland growth is compensating for stagnation or offsetting changes in the other components. The informative assessment states, at every level at which an aggregate is used: component endpoint ratios, log contributions, attribution shares, data vintage and endpoint sensitivity.

S7 The one-stock comparator

Earlier versions of this work developed a single-stock model that the two-land model supersedes. It is retained here as a comparator so that quantities that circulate for it are not mistaken for two-land results. The comparator has three states (A, P, D) and one algebraic relation:

$$G(A) = \rho A(1 - A/A_{\max}), \qquad B = bA + b_G G(A), \qquad E = eP, \tag{S5}$$

$$\dot{A} = G\big(A(t-\tau_g)\big) - \left[E - \sigma b A\right]_+ / b_G, \tag{S6}$$

$$\dot{P} = rP\big(1 - P/K(t-\tau_p)\big), \qquad K = B/e, \tag{S7}$$

$$\dot{D} = [E - B]_+ - \eta D, \qquad b = (b_0 + T_b)e^{-\alpha D}, \tag{S8}$$

Here A and $A_{\max}$ are comparator-specific stock units and $G(A)$ has units stock unit yr^{-1}; consequently $b, b_0, T_b, \Delta b$ are gha per stock unit and b_G is gha yr per stock unit, so bA and $b_G G(A)$ are each gha. B and E are annual GFN/NFA account amounts in gha, P and K are in cap, and e is gha cap^{-1}. The shares σ are dimensionless. Time and the delays are in yr; ρ, r, η and the one-stock logistic-wave steepness κ are rates in yr^{-1} (κ is distinct from the two-land κ_w); D is a leaky time-integral in gha yr, so $\dot{D}$ and ηD are gha; and α is gha^{-1} yr^{-1}. The figures share a numerical ordinate only for illustration; A, P, B and D retain distinct units.

With demand met first from the flow bA, the shortfall liquidates the standing stock at rate $1/b_G$. Its properties, each of which was re-derived under the two-land structure and found not to carry over, are the following.

- *MSY and fold.* $B(A)$ has an interior maximum (the MSY of Schaefer, 1954) only in the capital-dominated regime $b_G\rho > b$; a fixed liability above $B_{\max}$ produces a saddle-node, the catastrophic-shift structure of Scheffer et al. (2001). At the comparator baseline ($b_G\rho = 0.04 < b = 0.5$ in gha per comparator-specific stock unit) the fold is absent. In the two-land model Y_c is linear in A_c and $B_{eq}(A_c)$ is monotone; there is no fold in any regime (Proposition 2).

- *The equilibrium family $P = B(A)/e$ and $\det J = 0\,\mathrm{yr}^{-3}$.* Under full harvest ($\sigma = 1$) the comparator has a neutral continuum of equilibria on which the Jacobian is exactly singular. The two-land model has a one-parameter family in the *land allocation* (indexed by A_c on the no-conversion face, with $A_r/A_f = \eta_f/\mu$), not a free P, and no identically singular Jacobian.

- *The $+0.6$ eigenvalue and the two-gain Hopf classification.* The comparator's deficit-region linearization has a positive real root ($a_{11} = G'(A^*) + b/b_G \approx 0.608$–$0.625\,\mathrm{yr}^{-1}$ at its baseline) and a dimensionless comparator-specific two-gain rule $\chi_{1s} = q/(\rho - 2q)$ (here q is the comparator's feedback-rate coefficient in yr^{-1}, not the two-land quality index) selecting which delay destabilizes (τ_g-only for $\chi_{1s} > 1$, τ_p-only for $\chi_{1s} < 1$, with critical delays $\tau_m \approx 85.4\,\mathrm{yr}$ and $\tau_p \approx 231\,\mathrm{yr}$). None of this is a two-land quantity: the two-land deficit-side spectrum is real and negative ($-0.054\,\mathrm{yr}^{-1}$, $-1.0\,\mathrm{yr}^{-1}$) and constant along the branch.

- *The regeneration cliff and the masking window.* The comparator shows a recover/collapse boundary at $\tau_g \approx 18$–$20\,\mathrm{yr}$ and a productivity-illusion window of $\approx 5.4\,\mathrm{yr}$ ($A : 1.00 \to 0.86$ comparator-specific stock units while $B : 0.578 \to 0.647\,\mathrm{gha}$; the mask vanishes for initial deficits $\gtrsim 0.075\,\mathrm{gha}$). In the two-land model the boundary is a demand threshold insensitive to τ_g (Table S2) and the composition window is set by λ_1 and Δb_{conv} (paper, Section 3.1). The dimensionless recovery fractions $0.54 \to 0.52 \to 0.50$ for $\tau_g = 5 \to 25 \to 40\,\mathrm{yr}$ quoted in earlier versions were computed with an area-based intermediate formulation and are superseded by the identical-to-10^{-3} result of Table S2.

The four selected comparator figures reproduced here are numbered S1–S4. The broader historical supplementary figure set (S1–S15) remains preserved in the repository at `agent 2 productivity illusion/supplementary/FIGURES/`; only figures used by the current SI are included in this package.

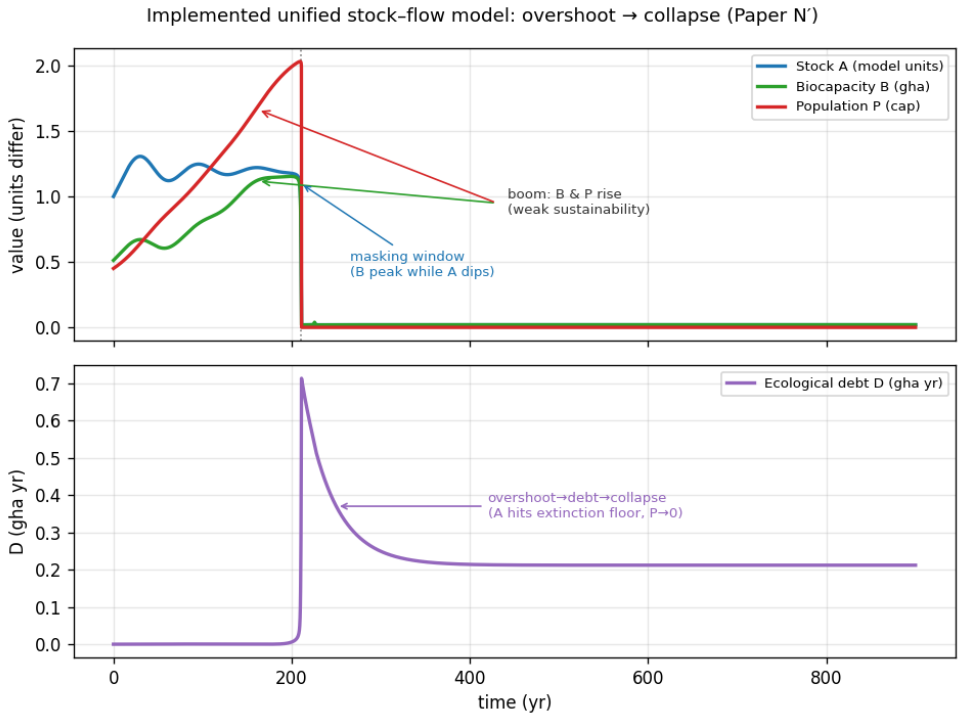


Figure S1: One-stock comparator: top panel shows stock A (model-specific stock units), population P (cap) and annual biocapacity B (gha); the lower panel shows leaky debt D (gha yr). The footprint $E = eP$ is a driver and is not plotted. The shared numerical ordinate is illustrative, not a common physical scale. Comparator only; not a two-land result.

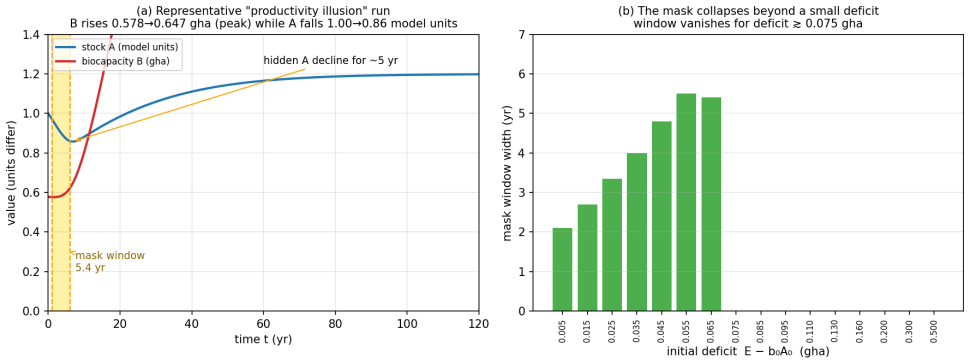


Figure S2: One-stock comparator: the productivity-illusion window (A falls from 1.00 to 0.86 model units while B rises from 0.578 to 0.647 gha; window ≈ 5.4 yr; the mask vanishes for deficits $\gtrsim 0.075$ gha). The two-land composition window of the paper (Fig. 1a) is a different object with a different mechanism.

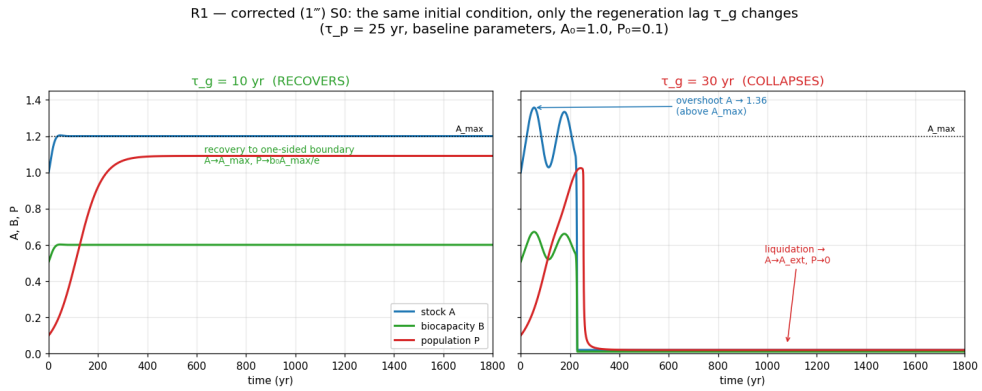


Figure S3: One-stock comparator: recovery versus collapse as a function of regeneration delay τ_g (yr; the $\approx$ 18–20 yr cliff). The plotted series use a shared numerical ordinate only for illustration: stock A is in comparator-specific units, biocapacity B in gha and population P in cap. Not present in the two-land model.

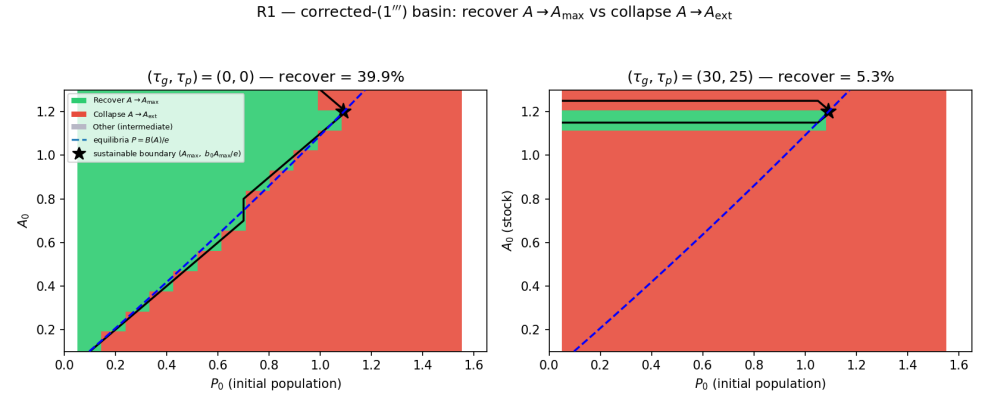


Figure S4: One-stock comparator: basin of recovery at the comparator baseline. The axes are initial population P_0 (cap) and initial stock A_0 (comparator-specific stock units); the equilibrium guide $P = B(A)/e$ is in cap, with annual account value B in gha and e in gha cap^{-1}. In the two-land model the corresponding object is the continuous drawdown-depth surface of Fig. 1c and Table S4.

S8 Controlled technology-wave experiment

This controlled extension tests whether an active T_b wave can produce a larger leaky-debt peak than an otherwise matched no-wave run. The pair uses endogenous demand $E = eP$ (so the surplus-gated R_{rc} is active only when $B > E$), $A_{tot} = 2.5\,\mathrm{ha}$, $A_{f0} = 0.6\,\mathrm{ha}$, $A_{c0} = 0.9\,\mathrm{ha}$, $q_0 = 0.9$, $D_0 = 0.0\,\mathrm{gha\,yr}$, and $P_0 = 1.0$ or $1.2\,\mathrm{cap}$. Shared parameters are $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$, $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$, $b_{G,c} = 0.8\,\mathrm{gha\,yr\,ha^{-1}}$, $\rho_c = 0.08\,\mathrm{yr^{-1}}$, $\alpha = 0.03\,\mathrm{gha^{-1}\,yr^{-1}}$, $\eta = 0.05\,\mathrm{yr^{-1}}$, $\eta_f = 0.05\,\mathrm{yr^{-1}}$, $\mu = 0.06\,\mathrm{yr^{-1}}$, $e = 0.55\,\mathrm{gha\,cap^{-1}}$, $r = 0.02\,\mathrm{yr^{-1}}$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $\tau_{conv} = 1\,\mathrm{yr}$, and $\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$. The wave parameters are $\Delta b = 1.0\,\mathrm{gha\,ha^{-1}}$, $\kappa_w = 0.1\,\mathrm{yr^{-1}}$, and $t_{wave} = 40\,\mathrm{yr}$; the horizon is $T = 250\,\mathrm{yr}$. Within each pair, the physical initial state, parameters, wave-free pre-run history, and solver are identical; only the T_b switch differs. The logistic is already nonzero at $t = 0$ ($T_b(0) = 0.017986\,\mathrm{gha\,ha^{-1}}$), so the derived initial account is $B(0) = 0.560184\,\mathrm{gha}$ without the wave and $0.570976\,\mathrm{gha}$ with it. This small initial difference is part of the declared forcing, not a change to A_f, A_c, q, P, or D; T_b reaches half its increment at the marked midpoint $t_{wave} = 40\,\mathrm{yr}$.

Figure S5 shows the $P_0 = 1.0$ pair. Its all-time debt maximum occurs after the wave midpoint in both runs: $D_{max} = 0.008126\,\mathrm{gha\,yr}$ at $86.15\,\mathrm{yr}$ without the wave, versus $0.020205\,\mathrm{gha\,yr}$ at $115.60\,\mathrm{yr}$ with it. At $P_0 = 1.2$, by contrast, the all-time maximum occurs early and is lower with the wave, while the maximum over $t \geq t_{wave}$ is higher. Table S13 keeps these all-time and post-midpoint definitions separate. We call $t \geq t_{wave}$ the post-midpoint window; it is not an assertion that a logistic wave has ended at t_{wave}. The full metrics and trajectories are deposited as `work/tb_wave_debt_metrics.csv`, `work/tb_wave_debt_timeseries_P0_1p0.csv`, and `work/tb_wave_debt_timeseries_P0_1p2.csv`; `work/tb_wave_debt_scan.py` regenerates them and the figure.

Table S13: Matched technology-wave/no-wave debt maxima. Entries are D_{max} in gha yr, with the time of the maximum in years in parentheses; the ratio is wave/no-wave. "All" is the maximum over $0 \leq t \leq 250\,\mathrm{yr}$; $t \geq 40$ is the post-midpoint maximum. The $P_0 = 1.0$ all-time peaks both occur after the midpoint.

P_0 (cap)	quality update	Δt (yr)	window	no wave D_{max} (time)	wave D_{max} (time)	ratio
1.0	delayed	0.050	all	0.008126 (86.15)	0.020205 (115.60)	2.487
1.0	delayed	0.025	all	0.008105 (86.20)	0.020195 (115.65)	2.492
1.0	current	0.050	all	0.008326 (85.70)	0.020168 (115.60)	2.422
1.2	delayed	0.050	all	0.061218 (1.65)	0.051037 (1.45)	0.834
1.2	delayed	0.050	$t \geq 40$	0.010854 (81.70)	0.025184 (112.55)	2.320
1.2	delayed	0.025	all	0.060732 (1.625)	0.050582 (1.475)	0.833
1.2	delayed	0.025	$t \geq 40$	0.010808 (81.775)	0.025165 (112.625)	2.328

The $P_0 = 1.0$ result is a declared sensitivity case, not a refit or a universal effect: wave/no-wave all-time peak ratios persist under step halving (2.487 at $\Delta t = 0.05\,\mathrm{yr}$; 2.492 at $0.025\,\mathrm{yr}$) and under the current-quality solver (2.422 at $0.05\,\mathrm{yr}$). At the reference $P_0 = 1.2$, the corresponding all-time ratios are 0.834 and 0.833, whereas the post-midpoint ratios are 2.320 and 2.328. All reported debt maxima change by less than 0.9% under step halving. Maximum area-conservation residual is $4.44 \times 10^{-16}\,\mathrm{ha}$; the minimum reserve in the $P_0 = 1.0$ wave run is $0.286\,\mathrm{ha}$, above the $0.10\,\mathrm{ha}$ floor. In this sensitivity pair A_c does not fall below its initial $0.9\,\mathrm{ha}$, so the experiment isolates a debt rebound and is not evidence of the capital drawdown shown in the main frozen-demand run. The regression test also confirms that an integer input $D_0 = 0$ still produces a floating-point debt state and finite debt accumulation. D is the model's leaky time-integral of shortfall, not an observed NFA or physical-stock variable. These conditional trajectories demonstrate a possible delayed rebound, while the $P_0 = 1.2$ contrast shows why the all-time peak cannot be substituted for the post-midpoint response.

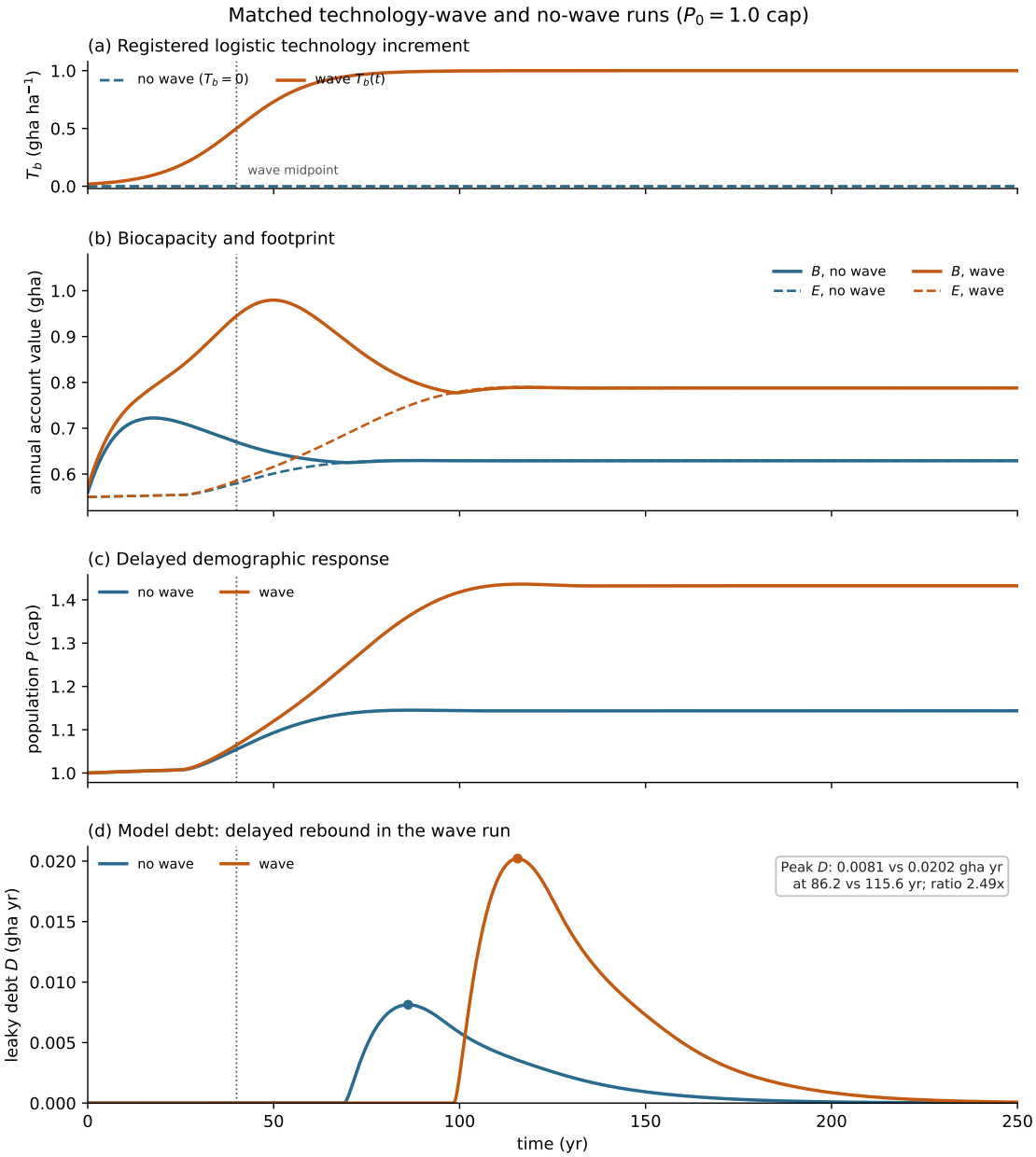


Figure S5: Matched endogenous-demand technology-wave/no-wave simulation at $P_0 = 1.0$ cap. Parameters and initial conditions are listed in SI §S8; only T_b is switched on or off. (a) The active logistic increment $T_b(t)$, with the dotted line at its midpoint $t_{wave} = 40$ yr; (b) annual account values B and E (gha); (c) population P (cap); (d) leaky model debt D (gha yr). Markers denote all-time maxima, which occur at 86.15 yr without the wave and 115.60 yr with it; peak debt is 0.008126 versus 0.020205 gha yr. The plotted wave begins with its small logistic tail at $t = 0$ and approaches $\Delta b = 1.0$ gha ha^{-1}; the dotted line marks the midpoint, not an end date.

Supplementary references

References

Abaee, A. (2026a). *Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter.* Zenodo. https://doi.org/10.5281/zenodo.22554217.

Abaee, A. (2026b). *Aggregate Indices and Transition Safety: A Quantifier-Order Separation Between Scalarized and Coordinate-Wise Feasibility.* Zenodo. https://doi.org/10.5281/zenodo.22545740.

Abaee, A. (2026c). *Typed Flux Ledgers and Depletion Arithmetic: Conservation, Componentwise Diagnostics, and the Semantics of Depletion Horizons.* Zenodo. https://doi.org/10.5281/zenodo.22554177.

Anderies, J. M. (2000). On modeling human behavior and institutions in simple ecological economic systems. *Ecological Economics*, 35(3), 393–412.

Becker, W., Saisana, M., Paruolo, P. & Vandecasteele, I. (2017). Weights and importance in composite indicators: Closing the gap. *Ecological Indicators*, 80, 12–22.

Blomqvist, L., Brook, B. W., Ellis, E. C., Kareiva, P. M., Nordhaus, T. & Shellenberger, M. (2013). Does the shoe fit? Real versus imagined ecological footprints. *PLoS Biology*, 11(11), e1001700.

Borucke, M., Moore, D., Cranston, G., Gracey, K., Iha, K., Larson, J., Lazarus, E., Morales, J. C., Wackernagel, M. & Galli, A. (2013). Accounting for demand and supply of the biosphere's regenerative capacity: The National Footprint Accounts' underlying methodology and framework. *Ecological Indicators*, 24, 518–533.

Brander, J. A. & Taylor, M. S. (1998). The simple economics of Easter Island: A Ricardo–Malthus model of renewable resource use. *American Economic Review*, 88(1), 119–138.

Calvin, K. & Bond-Lamberty, B. (2018). Integrated human–Earth system modeling — state of the science and future directions. *Environmental Research Letters*, 13(6), 063006.

Carvalhais, N., Forkel, M., Khomik, M., Bellarby, J., Jung, M., Migliavacca, M., Mu, M., Saatchi, S., Santoro, M., Thurner, M., Weber, U., Ahrens, B., Beer, C., Cescatti, A., Randerson, J. T. & Reichstein, M. (2014). Global covariation of carbon turnover times with climate in terrestrial ecosystems. *Nature*, 514(7521), 213–217.

Cohen, J. E. (1995). Population growth and Earth's human carrying capacity. *Science*, 269(5222), 341–346.

Copernicus (2019). *Global Land Cover LC100.* European Commission, Land Copernicus.

Dasgupta, P. (2021). *The Economics of Biodiversity: The Dasgupta Review.* London: HM Treasury.

Diamond, J. (2005). *Collapse: How Societies Choose to Fail or Succeed.* New York: Viking.

Donges, J. F., Winkelmann, R., Cornell, S. E., Lucht, W., Dyke, J. G., Rockström, J., Heitzig, J. & Schellnhuber, H. J. (2017). Closing the loop: Reconnecting human dynamics to Earth system science. *The Anthropocene Review*, 4(2), 151–157.

Erb, K.-H., Luyssaert, S., Meyfroidt, P., Pongratz, J., Don, A., Kloster, S., Kuemmerle, T., Fetzel, T., Fuchs, R., Hasegawa, T., Haberl, H., et al. (2017). Land management: data availability and process understanding for global change studies. *Global Change Biology*, 23(2), 512–533.

ESA (2017). *Land Cover CCI Product User Guide, Version 2.0.* European Space Agency.

Fischer, S. M., Joy, M. K., Abrahamse, W., Milfont, T. L. & Petherick, L. M. (2022). The use and misuse of composite environmental indices. *bioRxiv*, 2022.03.15.484501. https://doi.org/10.1101/2022.03.15.484501.

Global Footprint Network (2009). *Ecological Footprint Standards 2009.* Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/Ecological_Footprint_Standards_2009.pdf.

Global Footprint Network (2019). *Working Guidebook to the National Footprint Accounts 2019*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/National_Footprint_Accounts_Guidebook_2019.pdf.

Galli, A., Giampietro, M., Goldfinger, S., Lazarus, E., Lin, D., Saltelli, A., Wackernagel, M. & Müller, F. (2016). Questioning the ecological footprint. *Ecological Indicators*, 69, 224–232.

Giampietro, M. & Saltelli, A. (2014). Footprints to nowhere. *Ecological Indicators*, 46, 610–621.

Haberl, H., Erb, K.-H., Krausmann, F., Gaube, V., Bondeau, A., Plutzar, C., Gingrich, S., Lucht, W. & Fischer-Kowalski, M. (2007). Quantifying and mapping the human appropriation of net primary production in earth's terrestrial ecosystems. *PNAS*, 104(31), 12942–12947.

Henderson, K. & Loreau, M. (2018). How ecological feedbacks between human population and land cover influence sustainability. *PLoS Computational Biology*, 14(8), e1006389.

Hurtt, G. C., Chini, L. P., Frolking, S., Betts, R. A., Feddema, J., Fischer, G., et al. (2011). Harmonization of land-use scenarios for the period 1500–2100: 600 years of global gridded annual land-use transitions, wood harvest, and resulting secondary lands. *Climatic Change*, 109(1–2), 117–161.

Hutchings, J. A. & Reynolds, J. D. (2004). Marine fish population collapses: consequences for recovery and extinction risk. *BioScience*, 54(4), 297–309.

Hutchinson, G. E. (1948). Circular causal systems in ecology. *Annals of the New York Academy of Sciences*, 50(4), 221–246.

Klein Goldewijk, K., Beusen, A., Doelman, J. & Stehfest, E. (2017). Anthropogenic land use estimates for the last 12,000 years — HYDE 3.2. *Earth System Science Data*, 9(2), 927–953.

Krausmann, F., Erb, K.-H., Gingrich, S., Haberl, H., Bondeau, A., Gaube, V., Lauk, C., Plutzar, C. & Searchinger, T. D. (2013). Global human appropriation of net primary production doubled in the 20th century. *PNAS*, 110(25), 10324–10329.

Kuang, Y. (1993). *Delay Differential Equations, with Applications in Population Dynamics.* Academic Press.

Lin, D., Hanscom, L., Murthy, A., Galli, A., Evans, M., Neill, E., et al. (2018). Ecological footprint accounting for countries: updates and results of the National Footprint Accounts. *Sustainability*, 10(12).

Lo, K., Basnet, N., Miller, E., Srikanthalingam, B., Foley, B., Van Berkum, J. L., Long, A. H., Toneva, P., Ermina, M. & Klinkenberg, C. (2026). *National Ecological Footprint and Biocapacity Accounts, Edition 2026 (1961–2025): Results and metadata, version 1.0.* Prepared for Footprint Data Foundation by researchers at York University and the University of Iceland. Workbook and release notes: https://footprint.info.yorku.ca/data/.

May, R. M. (1973). *Stability and Complexity in Model Ecosystems.* Princeton: Princeton University Press.

Meadows, D. H., Meadows, D. L., Randers, J. & Behrens, W. W. (1972). *The Limits to Growth.* New York: Universe Books.

Meadows, D. H., Randers, J. & Meadows, D. L. (2004). *The Limits to Growth: The 30-Year Update.* White River Junction, VT: Chelsea Green.

Meadows, D. L., Behrens, W. W., Meadows, D. H., Naill, R. F., Randers, J. & Zahn, E. (1974). *Dynamics of Growth in a Finite World.* Cambridge, MA: Wright-Allen Press.

Meyer, P. S. (1994). Bi-logistic growth. *Technological Forecasting and Social Change*, 47(1), 89–102.

Meyer, P. S. & Ausubel, J. H. (1999). Carrying capacity: A model with logistically varying limits. *Technological Forecasting and Social Change*, 61(3), 209–214.

Monfreda, C., Ramankutty, N. & Foley, J. A. (2008). Farming the planet: 2. Geographic distribution of crop areas, yields, physiological types, and net primary production in the year 2000. *Global Biogeochemical Cycles*, 22(1), GB1022.

Motesharrei, S., Rivas, J., Kalnay, E., Asrar, G. R., Busalacchi, A. J., Cahalan, R. F., Cane, M. A., Colwell, R. R., Feng, K., Franklin, R. S., et al. (2016). Modeling sustainability: Population, inequality, consumption, and bidirectional coupling of the Earth and human systems. *National Science Review*, 3(4), 470–494.

Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffmann, A. & Giovannini, E. (2008). *Handbook on Constructing Composite Indicators: Methodology and User Guide.* Paris: OECD Publishing.

Neubauer, P., Jensen, O. P., Hutchings, J. A. & Baum, J. K. (2013). Resilience and recovery of overexploited marine populations. *Science*, 340(6130), 347–349.

Our World in Data (2026a). *Cropland area.* FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cropland-area.

Our World in Data (2026b). *Cereal yield.* FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cereal-yield.

Pearl, R. (1925). *The Biology of Population Growth.* New York: Alfred A. Knopf.

Poeplau, C., et al. (2011). Temporal dynamics of soil organic carbon after land-use change. *Global Change Biology*, 17(7), 2415–2427.

Poorter, L., et al. (2016). Biomass resilience of Neotropical secondary forests. *Nature*, 530(7590), 211–214.

Schaefer, M. B. (1954). Some aspects of the dynamics of populations important to the management of the commercial marine fisheries. *Bulletin of the Inter-American Tropical Tuna Commission*, 1(2), 25–56.

Scheffer, M., Carpenter, S., Foley, J. A., Folke, C. & Walker, B. (2001). Catastrophic shifts in ecosystems. *Nature*, 413(6856), 591–596.

Simon, J. L. (1996). *The Ultimate Resource 2.* Princeton: Princeton University Press.

Tainter, J. A. (1988). *The Collapse of Complex Societies.* Cambridge: Cambridge University Press.

United Nations et al. (2021). *System of Environmental-Economic Accounting—Ecosystem Accounting (SEEA EA).* White cover publication, pre-edited text subject to official editing. New York: United Nations. https://seea.un.org/ecosystem-accounting.

United Nations (2022). *World Population Prospects 2022: Summary of Results.* New York: Department of Economic and Social Affairs, Population Division.

van den Bergh, J. C. J. M. & Grazi, F. (2015). Reply to the first systematic response by the Global Footprint Network to criticism: a real debate finally? *Ecological Indicators*, 58, 458–463.

van den Bergh, J. C. J. M. & Rietveld, P. (2004). Reconsidering the limits to world population: Meta-analysis and meta-prediction. *BioScience*, 54(3), 195–204.

Wackernagel, M. & Rees, W. (1996). *Our Ecological Footprint: Reducing Human Impact on the Earth.* New Society Publishers.

Wackernagel, M., et al. (2002). Tracking the ecological overshoot of the human economy. *PNAS*, 99(14), 9266–9271.

Wilson, E. O. (2016). *Half-Earth: Our Planet's Fight for Life.* New York: Liveright.